

Harsha Kumar M R

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Professional Summary

Results-driven final-year Computer Science Engineer (CGPA: 9.03/10) with hands-on industry experience as an SDE Intern building production-grade SaaS platforms. Proficient in full-stack development (MERN), RESTful API design, system design, and machine learning. Published IEEE researcher with a track record of delivering scalable, secure, and high-performance software. Seeking a Software Development Engineer role to architect robust backend systems and drive impactful product features.

Technical Skills

Languages: Python, JavaScript, Java, C, SQL

Backend: Node.js, Express.js, FastAPI, RESTful APIs, JWT Authentication, Microservices

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Responsive Design

Databases: MongoDB, MySQL, Database Design, Data Modeling, Query Optimization

ML / AI: TensorFlow, Keras, Scikit-learn, NumPy, Pandas, CNN, SHAP, Computer Vision

Tools & Platforms: Git, GitHub, Postman, VS Code, bcrypt, Linux

Core CS: Data Structures & Algorithms, OOP, System Design, Operating Systems, DBMS, Agile/Scrum

Experience

Software Development Engineer Intern

Jan 2026 – Present

Wimera Systems Pvt Ltd — IoT-based Industrial SaaS, Bengaluru

- Engineered backend modules for production planning (process creation, job orders, work order workflows) on a live SaaS platform serving smart factory clients, reducing manual paperwork by digitizing end-to-end factory operations.
- Architected a 3D factory visualization system with real-time IoT machine-status updates using color-coded indicators, improving operator response time for anomaly detection.
- Designed and deployed RESTful APIs consumed by the React.js frontend, maintaining sub-200 ms response times in a production environment.
- Collaborated cross-functionally on deployment and feature scaling; followed Agile sprint cycles with regular code reviews.
- Tech Stack:** Node.js, Express.js, MongoDB, React.js, REST APIs

Education

Global Academy of Technology

Aug 2022 – May 2026

B.E. Computer Science and Engineering — **CGPA: 9.03 / 10**

Bengaluru, Karnataka

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Software Engineering, Machine Learning, Web Technologies, Computer Networks

BGS PU College

Jun 2020 – Mar 2022

PUC (PCMB) — **94.83%**

Magadi, Karnataka

Projects

Malicious SQL Query Detection System — Hybrid CNN + Rule-Based Model

2025

- Designed a hybrid CNN + rule-based architecture achieving **95%+ accuracy** at **<100 ms latency** for real-time SQL injection detection.
- Applied adversarial training with temporal data splits, reducing model overfitting by improving generalization across unseen attack patterns.
- Integrated SHAP-based explainability to surface interpretable security analytics for analysts.
- Built a full-stack monitoring interface (React.js + FastAPI) with live query ingestion, result visualization, and configurable alert generation.

- **Tech Stack:** Python, TensorFlow, Keras, FastAPI, React.js, MySQL, Pandas, SHAP

Weather Analytics Dashboard — Full-Stack Web Application

2025

- Developed a full-stack application with real-time weather data, hourly trend visualization, and 7-day forecasts for any global city.
- Implemented OAuth 2.0 Google Sign-In, personalized user preferences with persistent storage, and fully responsive UI.
- Built RESTful Node.js/Express.js backend integrating OpenWeather API; optimized API calls to reduce latency.
- **Tech Stack:** MERN Stack, OpenWeather API, JWT, OAuth 2.0

Online Bus Ticket Booking & Reservation System

2024

- Engineered a scalable MERN-stack platform supporting real-time seat reservation, booking management, and role-based admin controls.
- Implemented JWT-based authentication with bcrypt password hashing, ensuring secure multi-user access.
- **Tech Stack:** MongoDB, Express.js, React.js, Node.js, JWT, bcrypt

Role-Based Task Distribution System

2024

- Built a role-based task management platform with automated CSV-driven task distribution, reducing manual assignment overhead.
- Implemented RBAC with JWT security and real-time dashboard monitoring for managers and team leads.
- **Tech Stack:** MERN Stack, JWT, CSV Processing

Publications

Fingerprint Liveness Detection Using Deep Learning

Co-authored and presented at the **IEEE CIISCA Conference** — applied CNN-based classification for biometric spoofing detection.

Certifications

- **NPTEL** — Programming, Data Structures and Algorithms using Python
- **Infosys Springboard** — Cloud Computing (Azure), Linux, Python, Machine Learning, Agile Scrum

Achievements & Leadership

- **Flipkart GRiD 6.0 (2024):** Developed a scalable e-commerce prototype with optimized product search and cart workflows; advanced to national rounds.
- **Hack-A-League 3.0 (2025):** Built an AI-powered workflow automation prototype using NLP pipelines; recognized as hackathon finalist.